

PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT 107	K.L. Man & G. Mogos	CPT	1509

2023/24 SEMESTER 1 – Final Exam

BACHELOR DEGREE – Year 2

Discrete Mathematics and Statistics

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. Total marks available are 100, accounting for 80% of the overall module marks.
2. The number in the column on the right indicates the marks for each question.
3. Answer all questions.
4. Answers should be written in English and in the answer script provided.
5. Relevant and clear steps should be included in your answers.

Notes:

- To obtain full marks for each question, relevant and clear steps need to be included in the answers.
- Partial marks may be awarded depending on the degree of completeness and clarity.

Question 1: Proof Techniques

[12 marks]

- (a) Use proof by *contradiction* to show that there does not exist any integers a and b such that $5a + 15b = 1$.
(2 marks)
- (b) Let $x \in \mathbb{Z}$. If $x^2 - 4x + 7$ is even, then x is odd. If you think that it is true, prove it. Otherwise, give a counterexample.
(5 marks)
- (c) Let $n \in \mathbb{Z}$ and $n \geq 1$, use proof by induction to show that:

$$3(1^2 + 2^2 + 3^2 + \dots + (2n)^2) = n(4n + 1)(2n + 1).$$

(5 marks)

Question 2: Set Theory

[14 marks]

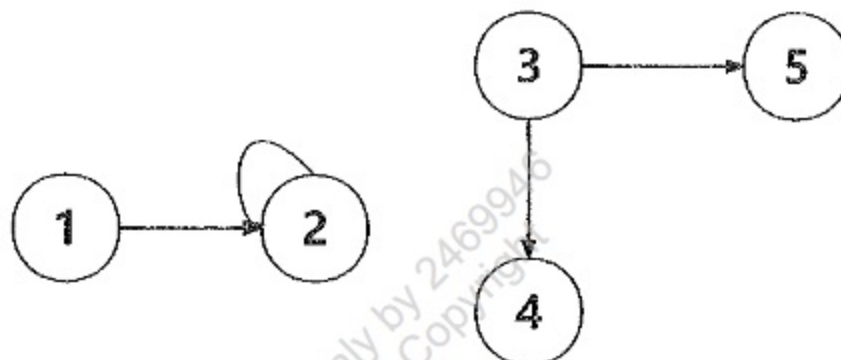
- (a) Let the universal set be $\mathbb{Z}$, $X = \{x \in \mathbb{Z} \mid x \text{ is odd and } 2 < x < 8\}$,
 $Y = \{y \in \mathbb{Z} \mid y \text{ is even, } y \neq 0, y \neq 4 \text{ and } 2 < y < 10\}$,
 $Q = \{q \in \mathbb{Z} \mid q \text{ is odd, } 1 < q < 17 \text{ and } 3q = 45\}$ and $P = \{p \in \mathbb{Z} \mid p^2 + 3p - 40 = 0\}$.
Find the elements of the following expressions:
1. $(X \cup Q) \cap P$
 2. $Y \cup (Q - X)$
 3. $(Y \cap \sim X) \cap (X \Delta Q)$
 4. $((X \cup Q) \cap Y) - ((Y \cap \sim X) \cap (X \Delta Q))$
- (4 marks)
- (b) If Q and P are empty sets, then $Q = P$. If you think that it is true, prove it. If not, explain why.
(4 marks)
- (c) Prove the following set identity using set theory properties and operations:
 $(A \cup (B \cap C)) = (A \cup B) \cap (A \cup C)$
(6 marks)

Question 3: Relations

[22 marks]

- (a) Which relation is represented by the following digraph?

(2 marks)



- (b) If M and N are both transitive relations, then their intersection is also transitive. If you think that it is true, prove it. If not, give a counter-example.

(4 marks)

- (c) Let $P = \{1, 2, 3, 4\}$. What is the transitive closure of the relation $\{(1, 2), (2, 3), (3, 4), (1, 3), (1, 4), (2, 4)\}$ on P ? Express your answer using ordered pairs.

(4 marks)

- (d) Determine whether or not the following relation is a partial order or even a total order.

 $(\mathbb{Z} \times \mathbb{Z}, \preceq)$ such that $(a, b) \preceq (c, d)$ if $(a \leq c)$ and $(b \geq d)$.

(6 marks)

- (e) Prove or disprove that the transitive closure of R^2 must be the transitive closure of R , where R be a relation on Z .

(6 marks)

Question 4: Functions and PHP**[15 marks]**(a) Consider the function $f: [0, +\infty) \rightarrow [3, +\infty)$ given by $f(x) = 2x + 3$.

- (1). Prove that f is injective. (3 marks)
- (2). Prove that f is surjective. (3 marks)
- (3). Find the inverse of f . (3 marks)

(b) Let $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = x + 2a, \quad g(x) = 2x + a \quad \text{and} \quad h(x) = x + 5, \quad \text{with } a \in \mathbb{R}.$$

Find a if $h \circ (g \circ f) = (f \circ g) \circ h$.

(3 marks)

(c) Tom sorts out his building blocks according to size. In a box he has building blocks that have the same shape and size but different colours. In one of the boxes, he has 50 red blocks, 35 blue, 25 green and 40 white.

What is the minimum number of blocks he can pick from this box to guarantee that he has picked an identical pair of blocks?

(3 marks)

Question 5: Logic**[22 marks]**

(a) Show whether the statements below are tautologies and/or contradictions:

1. $p \wedge (\neg p \vee q)$ (2 marks)
2. $(p \rightarrow q) \vee (p \rightarrow \neg q)$ (2 marks)
3. $(p \wedge q) \vee (\neg p \vee (p \wedge \neg q))$ (2 marks)

(b) For each of the two cases 1. and 2. below, show whether the statement on the left-hand side is equivalent to the statement on the right-hand side:

1. $\neg(p \rightarrow q) \equiv p \wedge \neg q$. (2 marks)
2. $(p \wedge q) \rightarrow (\neg r \vee \neg s) \equiv (r \wedge s) \rightarrow (\neg p \vee \neg q)$. (2 marks)

(c) Translate the following predicate formulas into English:

1. $\forall x[\text{Bag}(x) \rightarrow \exists y[\text{Coin}(y) \wedge \text{Contains}(x, y)]]$ (2 marks)
2. $\exists x \exists y[\text{IsSisterOf}(x, \text{tom}) \wedge \text{IsBrotherOf}(y, \text{mary}) \wedge \text{Knows}(x, y)]$ (2 marks)

(d) Let S be the signature consisting of a unary predicate symbol $Film$, a binary predicate symbol $IsActorOf$ and the individual constants $brad$ and $babylon$. Assume that these symbols have the following meaning:

- $Film(x)$ states: x is a film.
- $IsActorOf(x, y)$ states: x is an actor in y .
- $brad$ and $babylon$ refer to "Brad" and "Babylon", respectively.

Translate the following sentences into S-formulae:

1. "Babylon" is not a film.
2. Brad is an actor in "Babylon".
3. There is a film that Brad is an actor in.
4. Brad is an actor in at least 2 films.

(8 marks)

Question 6: Combinatorics and Probability

[15 marks]

(a). A delegation of 4 students is selected each year from a college to attend the National Student Association annual meeting. There are 12 eligible students.

- 1) In how many ways can the delegation be chosen?
- 2) In how many ways if two of the eligible students will not attend the meeting together?
- 3) In how many ways if two of the eligible students are married and will only attend the meeting together?

(6 marks)

(b). Box A contains 8 items of which 3 are defective, and box B contains 5 items of which 2 are defective. An item is drawn at random from each box.

- (1) What is the probability p that both items are non-defective? (2 marks)
- (2) What is the probability p that one item is defective and one not? (2 marks)
- (3) If one item is defective and one is not, what is the probability p that the defective item came from box A? (2 marks)

(c) Your company plans to invest in a particular project. There is a 35% chance that you will lose 30,000 yuan, a 40% chance that you will break even, and a 25% chance that you will make 55,000 yuan. Calculate the expected value of the project.

(3 marks)

END OF EXAM PAPER

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Notes:

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- Partial marks may be awarded depending on the degree of completeness and clarity.

Question 1: Proof Techniques

[12 marks]

- (a) Use proof by *contradiction* to show that there does not exist any integers a and b such that $7a + 21b = 1$.

(2 marks)

- (b) Let $x \in \mathbb{Z}$. If $x^2 - 4x + 9$ is even, then x is odd. If you think that it is true, prove it. Otherwise, give a counterexample.

(5 marks)

- (c) Let $n \in \mathbb{Z}$ and $n \geq 1$, use proof by induction to show that:

$$2 + 5 + 8 + \dots + (3n - 1) = \frac{n}{2} (3n + 1).$$

(5 marks)

Question 2: Set Theory

[16 marks]

- (a) Let the universal set be $\mathbb{N}$, $X = \{x \in \mathbb{N} \mid x \text{ is odd and } 2 < x < 10\}$, $Y = \{x \in \mathbb{N} \mid x \text{ is even and } 1 < x < 7\}$ and $Z = \{x \in \mathbb{N} \mid x \text{ is odd and } 2 < x < 8\}$. Find the elements of:

1. $X \cap (Y \cup Z)$
2. $\sim((X - Z) \cap Y)$
3. $(Z \cup X) \cap (X \Delta Y)$
4. $((Z \cup X) \cap (X \Delta Y)) - (X \cap (Y \cup Z))$

(4 marks)

- (b) If $X \subset Z$ and $X \subseteq Y \subseteq Z$, then either $X \subset Y$ or $Y \subset Z$. If you think that it is true, prove it. If not, explain why.

(4 marks)

- (c) Let A and B be sets, and $A \subseteq B$ if and only if $(B - A) \cup A = B$. If you think that it is true, prove it. Otherwise, give a counterexample to show that it is false.

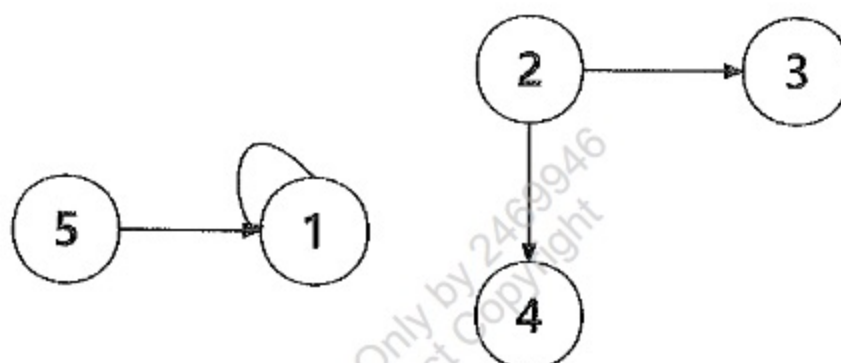
(8 marks)

Question 3: Relations

[18 marks]

(a) Which relation is represented by the following digraph?

(2 marks)

(b) Let $P = \{1, 2, 3, 4, 5, 6\}$. What is the binary relation on P represented by the below matrix?

$$\begin{pmatrix}
 0 & 1 & 0 & 1 & 0 & 0 \\
 0 & 0 & 1 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 1 & 0 \\
 0 & 1 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 1 & 0
 \end{pmatrix}$$

(4 marks)

(c) Let $A = \{a, b, c, d\}$. What is the transitive closure of the relation: $\{(a, c), (a, d), (b, a), (c, b)\}$ on A ?

(4 marks)

(d) Let $x R y$ on the set $A = \{2, 3, 6, 12, 24, 36\}$ whenever y is divisible by x . Prove or disprove R is a partial order. If R is a partial order, represent R as a Hasse diagram. Otherwise, prove R is not total order.

(8 marks)

Question 4: Functions and PHP**[15 marks]**

(a) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = (x+2)^2 - 3$ and $g(x) = ax + b$ with a, b , real constants.

Find the value of $a, a > 0$ and the corresponding value of b , such that:

$$(f \circ g)(x) = f(g(x)) = 4x^2 + 6x - \frac{3}{4}$$

(6 marks)

(b) Let $f: \mathbb{R}/\{\frac{4}{3}\} \rightarrow \mathbb{R}$ be given by $f(x) = \frac{x+7}{3x-4}$. Find $f^{-1}(x)$ and $f(f^{-1}(x))$.

(6 marks)

(c) A cinema has a seating capacity of 750. How many seats must be filled to ensure that at least two people seated in the cinema have the same first and last name initials?

(3 marks)

Question 5: Logic

[22 marks]

(a) Show whether the statements below are tautologies and/or contradictions:

1. $p \Leftrightarrow (p \vee p)$ (2 marks)
2. $(p \wedge \neg q) \wedge (\neg p \vee q)$ (2 marks)
3. $(q \wedge \neg(p \rightarrow q))$ (2 marks)

(b) For each of the two cases 1. and 2. below, show whether the statement on the left-hand side is equivalent to the statement on the right-hand side:

1. $p \rightarrow q \equiv \neg p \vee q$. (2 marks)
2. $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \equiv p \wedge q$. (2 marks)

(c) Translate the following predicate formulas into English:

1. $\exists x[\text{Mushroom}(x) \wedge \text{Ate}(\text{jane}, x) \wedge \text{Picked}(\text{jane}, x)]$ (2 marks)
2. $\forall x(\text{Dog}(x)) \rightarrow \exists y[\text{Person}(y) \wedge \text{Owns}(y, x)]$ (2 marks)

(d) Consider the signature $S = \{\text{Student}, \text{Male}, \text{Female}, \text{rebecca}, \text{helen}, \text{john}\}$ consisting of a binary predicate symbol *Student*, two unary predicate symbols *Male* and *Female*, and three constant symbols *rebecca*, *helen* and *john*. Assume that these symbols have the following meaning:

- *Student* means “is a student of” (i.e., *Student*(a, b) states a is a student of b).
- *Male* means “is male” (i.e., *Male*(a) states a is male).
- *Female* means “is female” (i.e., *Female*(a) states a is female).
- *rebecca*, *helen* and *john* refer to “Rebecca”, “Helen” and “John”, respectively.

Translate the following sentences into S-formulae:

1. Rebecca is a student of Helen.
2. John has a female student.
3. All students of John are also students of Helen.
4. Rebecca has at least 2 male students.

(8 marks)

Question 6: Combinatorics and Probability**[17 marks]**

(a) A student is to answer 8 out of 10 questions in an exam.

- 1) How many choices does he have? (2 marks)
- 2) How many choices does he have if he must answer the first 3 questions? (2 marks)
- 3) How many choices does he have if he must answer at least 4 of the first 5 questions? (2 marks)

(b) We roll two fair 6-sided dice. Each one of the 36 possible outcomes is assumed to be equally likely.:

- 1) Find the probability that doubles are rolled. (2 marks)
- 2) Find the probability that at least one die roll is a 6. (2 marks)
- 3) Given that the roll results in a sum of 4 or less, find the conditional probability that doubles are rolled. (2 marks)

(c) You pay 10 yuan to play the following game of chance.

There is a bag containing 12 balls, five are red, three are green and the rest are yellow. You are to draw one ball from the bag. You will win 14 yuan if you draw a red ball, 12 yuan if you draw a yellow ball and 0 yuan if you draw a green ball.

How much do you expect to win or lose if you play this game 100 times?

(5 marks)

END OF RESIT EXAM PAPER